

THIS IS BROUGHT TO YOU FOR THE PREPARATION OF CVE 105 EXAM

LECTURE ONE

1. When is principia mathematica published Ans 1687
2. The love of wisdom can be described as the desire to find out what is _____ and _____ Ans Real and True
3. Philosophy is a kind of _____ Ans Calling / Vocation
4. The definition of the term philosophy remains a subject of _____ Ans Controversy
5. The investigation of ultimate reality is _____ Ans Metaphysics
6. The study of morality and good is _____ Ans Ethics
7. _____ is the study of origin, validity and limits of knowledge. Ans Epistemology
8. _____ is the study of nature of beauty and art Ans Aesthetics
9. _____ is the arrangement of concept into a unified system Ans Synthetic Philosophy
10. _____ is the logical study of concept Ans Analytic Philosophy
11. The philosophy of science is equally, roughly speaking concerned with the nature of _____ Ans Scientific fact
12. The philosophers concern is not with observation and experiment but rather with the _____ of observation and experiment with the manner of interpretation. Ans Significance
13. The interpretation of observation leads to _____ Ans Knowledge
14. The volume of all gases is halved by doubling the _____ Ans Pressure
15. Newton laws of motion have been the subject of debate on and off ever since the publication of _____ Ans Principia Mathematica
16. Newton first law is also known as Ans Law of inertia
17. Name 5 philosophers that are the most important figures before 20th century
Descartes, John Locke, David Hume, Immanuel Kant, John Stuart mill Ans Aristotle, Rene
18. In the study of science, the composition of the universe is divided into two entities, Ans Matter and Energy
19. Matter include the material in the _____ Ans Universe
20. Energy can be measured only while being Ans Transformed
21. In what year did Albert Einstein express the relationship between matter and energy 1905
22. _____ is a systematic way of acquiring knowledge about a particular field of study Science
23. Science is roughly speaking concerned with what is concerned as _____ Fact
24. _____ is the product, tools and processes used to accomplish task on a daily bases Technology
25. Technology is the application of _____ to solve a problem Science
26. _____ has a goal of creating and improving artefacts and system to satisfy human want. Technology
27. _____ is an effort to reconstruct the past to discover what people thought, what they did and how their beliefs and action continue human want. History
28. Historian examine _____, first accountant hand of the people who lived though event. Primary Source
29. Primary source is often _____ or _____ Contradictory or Incomplete interpret
30. Historians _____ primary source evidence Secondary source
31. The story told by historians is known as Rewrites
32. Every generation of historians _____ the past Scientific method
33. _____ is a systematic way of learning about the world around us and answering questions. Hypothesis
34. _____ is a statement used to predict the outcome of an experiment. Hypothesis
35. _____ is a source of educated guess about what you expect Null hypothesis
36. _____ is an easy type of hypothesis Dependent and Independent
37. An experiment has a/an _____ and _____ variable Fact
38. Historians establish what they think as the _____ Philosophy
39. _____ is the investigation into the general nature of scientific practice Hands
40. The word technology relates to using of _____ in practical actions Solar panel
41. A good example of technology is the production of _____ Solar powered calculator
42. Solar panels are used in a variety of different technologies, but the simplest example is a Heat energy and light energy
43. It was proposed long ago that the sun emits _____ and _____ energy Engineer
44. In a sense, an _____ is one who makes an engine Engine
45. _____ is a mechanical device that enables man to perform task beyond his manual and physical strength Philosophy
46. _____ is the love of wisdom Real and True
47. The love of wisdom can describe as the desire to find out what is _____ and _____ Philosophy
48. _____ are used in the variety of different technologies Philosopher of science
49. The term _____ is often popularly used to indicate a set of basic values and attitudes History of Science and Technology
50. The business of the _____ is not to discover new facts about the world, but rather to take a critical look at what is claimed to be fact by the scientist. HST
51. What is the full meaning of HST?
52. _____ is a field in history which examines how humanity's understanding of the natural world (Science) and ability to manipulate it (Tech) have changed over the centuries.

LECTURE 2

- Science may be defined in terms of either _____ and _____ Knowledge and Method
- Scientific investigation may begin in response to _____ made by scientist or in response to some internal _____ process on the part of scientist. Observations / inductive
- Karl Popper stated that be capable of _____ Refutation
- The fact of scientific knowledge must be _____ & _____ by another scientist Testable and Repeatable
- It is essential that all scientific investigations are described _____ and _____ Fully and Clearly
- If investigation yield identical result under identical conditions, then the result may be accepted as _____ Valid
- Knowledge which cannot be investigated is not _____ and described as _____ Scientific, Metaphysical
- Fact are based on _____ obtained directly or indirectly by the senses. Observations
- All fact related to a particular problem are called Data
- Data can be values derived from _____, most often expressed in _____ form. Scientific experiment, Numerical
- The word data oriented as plural of the Latin word _____ meaning _____ Datum, something given
- Scientist use data to test what has been _____ in nature. Observed
- Observations may be _____ or _____ Qualitative/ Quantitative
- The _____ is more precise form of observation and involves the measurement of an amount or quantity. Latter
- _____ is obtained fundamentally by careful observation of the behavior of nature. Scientific knowledge
- A particular natural event is called a Phenomenon
- Hypothesis are scientific _____ Hunches
- Scientist use _____ to formulate a logical explanation for a phenomenon Hypotheses
- Scientist state hypotheses so they can test the _____ Hunch
- _____ are neither right nor wrong. Hypothesis
- Hypothesis can be tested by _____ Experiments
- When the weight of the evidence seems to indicate that hypothesis is sound, he then announces his findings as _____. If his theory is proven to be a non-varying performance, he said to have discovered a natural _____. Theory, Law
- It is much more common for teams of skilled _____ to work together on a single project. The work of these teams is called _____ Scientist, Research
- It is significant, the attitude of scientist is that no _____ or _____ represent absolute truth. Theory / Law
- Scientist feel, rather that each new _____ has brought man nearer to the truth Discovery
- _____ is the starting point that leads to new scientific discovery Observation
- Observation serve as a _____ for undertaking the study of field of science. Spring-board
- _____ is the application of scientific method Experiment
- Name the 4 parts of experiment. Problem, Procedure, Observation, Conclusion
- State 3 characteristics of scientific theory Rational, Relevant, Extensible
- A scientific theory that remains unchanged for years is known as a _____ Scientific Law
- _____ is a major constituent of scientific method Expert observation
- "Either the proper evidence has not yet been uncovered, or our present method are not yet capable of finding the required evidence" Carr / 1992
- _____ identified preconception as an initial state idea McCormack
- _____ is defined as a particular tendency or inclination, especially on that prevent impartial consideration of a question. Ans Bias
- Philosophy is the _____ and _____ into basic principles
- An important feature of an experiment is the _____ Control
- _____ and _____ group has the same diet, housing and activities Experimental and Control group.
- _____ describes color, shape, taste, presence and so on.
- _____ is obtained directly or indirectly by the sense of instruments.
- _____ may be used to seek an answer to a problem or test hypothesis. Experiment

LT 3

- _____ was discovered so that food can be cooked rather than eaten raw or dried Fire
- _____ resulted in the sudden stoppage of wandering and periodic supply of food Soil cultivation
- _____ is the earliest period of human culture when tools were made of stone, bones, antler or wood Stone age
- _____ is the period when man hunt animal flesh and gather fruits as food, animal skin and plant as clothes plant and physical shade as shelter. Normadic and Post Normadic era
- _____ was man first tool followed by _____, _____, _____, _____. Bare hands, Wood, Bones, Stone, Metals.
- The term _____ encompasses almost all the whole of human existence. Stone age
- Stone age was eventually subdivided by _____ and _____ in _____ year. British Naturalist, Politician John Sir Lubbock, 1865
- Stone age was subdivided into _____ and _____ with their meanings. Neolithic (New Stone), Palaeolithic (Old Stone)
- _____ was primary material for tool making in Stone age Flint

10. Flint is mostly used for _____ and _____ Tools and Weapons
11. The _____ tool were used for sharpening of wood and the _____ was used in harvesting during the late stone age. Adzes, Sickle
12. The early farming tool date from about _____ BC. 6000
13. The _____, bottom was used for clearing, _____, left is used for harvesting of cereal crops, a _____ and _____, center are used for grinding flour, the _____, upper right are used for ventilation of bread oven. Axe, Sickle, A flat rock rounded stone, perforated clay slab.
14. _____ is the period in antiquity that corresponds to the introduction of metallurgy. Bronze age
15. _____ are used in making tools, weapons and ceremonial objects. Bronze
16. In Europe, the bronze age spans the period of about _____ to _____ BC 2000 to 700
17. _____ is used in handling hot metals Tongs
18. The _____ age weapons include slender, spear heads, sword and knives Bronze
19. _____ denotes the period between end of bronze and spread roman empire. Iron Age
20. _____ was used in the manufacture of axe, saw hoes and nails. Iron Age
21. _____ is used to decorate personal items like mirror and pins. Bronze
22. _____ is the first true mathematician Pythagoras
23. _____ established a movement in 6th century BC southern Italy that emphasized the study of mathematics as a means to understanding all relationships in the natural world. Pythagoras
24. _____ are the followers of movement in Pythagoras. Pythagoreans
25. _____ were the first to teach the earth is a sphere revolving around sun Pythagoreans
26. _____ was a Greek philosopher and teacher who died in Athens around 400BC Socrates
27. _____ was the first to use the term philosophy Plato
28. Philosophy means _____ Love of knowledge
29. Plato was born around _____ BC 428
30. Aristotle was born at _____ Stagira
31. At what age did Aristotle became a pupil of Plato at Athens Seventeen
32. Aristotle taught that all things are made up of 4 elements namely Air, Earth, Fire, Water.
33. Aristotle writing covers the whole area of _____ Knowledge
34. Aristotle residence in Asia minor was _____ to _____ BC. 347, 342
35. The whole Aristotle's science was deeply influenced by his _____ experience. Biological
36. _____ studied the behavior of falling bodies and formulated law describing this behavior Galileo Galilei
37. _____ discovered that the orbits of the planet were elliptical in shape Johannes Keplers
38. _____ law is used in calculating the flight path of today's space vehicle. Keplers
39. _____ developed the law of motion and gravitation Isaac Newton
40. _____ made the discoveries about the nature and composition of light. Isaac Newton
41. _____ and _____ did much of pioneer work in electricity. Benjamin Franklin, Micheal Faraday
42. _____ and _____ put the study of heat on a Scholarly basis Count Rumford, James P. Joule
43. The industrial revolution started in the _____ century 19th
44. _____ remains the hallmark of the modern technological civilization Industrial Revolution
45. The _____ and _____ system of the era contributed principally to the great achievements of the 19th century. Economy, Social
46. The first person to use high pressure steam to power a vehicle was _____ in year _____. Richard Trevithick, 1801
47. The first engineers to usher in the rail way era were father and son _____ and _____ in year _____. George Stephenson, Robert Stephenson, 1855.
48. By 1855, there were over _____ km of track. 12,870
49. _____ in year _____ developed the new process for converting crude iron into steel. Henry Bessemer, 1855
50. The introduction of steam engine led to many new inventions in _____ and _____ Transport and Industries
51. Steam engine were built as early as _____ 1690
52. _____ arrived at the design of modern steam engine. James Watt.
53. The engraving of a steel factory illustrates the process developed by Sir. _____ in year _____. Henry Bessemer, 1855
54. The early 20th century, attention move to space exploration pioneered by Russian by the name of _____ and an American called _____ and a German man called _____ Konstantin K. Tsiolkovsky, Robert H. Goddard, Herman Oberth
55. The soviet launched its first satellite named _____ in October _____ around the earth orbit. Sputnik 1, 1957
56. The Russian Sputnik 1 was launched in October 4 _____ 1957
57. What is the full meaning of NASA and was created in _____ in October _____? United states, 1958
58. How many men walked on moon surface and returned to earth. 12
59. _____ and _____ manned the first space vehicle in the year _____. Soviet Cosmonaut Yury A. Gagarin, Alan B. Shephard Jr.
60. Alan B. Shephard Jr is _____ astronaut American
61. _____ and _____ became the first human to set foot on the moon and in what year. Neil Armstrong and Edwin E. Aldrin
62. Neil Armstrong and Edwin E. Aldrin are _____ astronauts. U.S.
63. _____ was utilized militarily during 2nd world war. Atomic energy

64. _____ country drop atomic bomb on Japanese cities. In what year and over how many people were killed. United State of America, 1945, 200,000
65. The disastrous effect of _____ nuclear power plant in the then USSR in _____ year. Chernobyl, 1986
66. Computer was first recognized when _____ and _____ were used as the first analog computing device. Stick, Ropes
67. _____ is the first computing instrument for exclusive general calculation was developed. ABACUS
68. Mechanical computing machine was invented by a German called _____ in year _____. Wilhem Schickard, 1623
69. _____ invented slide rule in _____ year. William Oughtred, 1930's
70. Second mechanical calculator was invented by a French scientist and a philosopher named _____ in year _____. Blaise Pascal, 1642
71. The difference engines were built by an English inventor named _____ in year _____. Charles babbage, 1822
72. Charles babbage built the digital analytical engine in _____ year. 1834
73. _____ built the electronic mechanical machine in what year. Herman Hollerith, 1880
74. Computer has been used to develop basic skill such as _____ and _____. Reading, Number work
75. Discuss about 5 famous scientist you know
76. Name the 8 development of science and technology and discuss.
77. _____ brought about rough division of labor where man does the heavier work, wife and children does the lighter domestic work. Normadic and Post Normadic era.

LT4

1. _____ is the remains of an animal or plant preserved from an earlier era inside a rock or geological deposit. Fossils
2. _____ also include the imprints of organisms that has dissolved. Fossils
3. _____ is a man-made object taken as a whole. Artefact
4. _____ is the study of life in prehistoric times by using fossil evidence. Paleontology
5. _____ is the study of fossils. The earth science that studies fossils organisms and related remains. Paleontology
6. _____ is an object made by human being, especially one that has archaeological or cultural interest. Artefacts
7. _____ is the sequence event involved in the evolutionary development of a spices or taxonomic group of organisms. Phylogeny
8. Human beings are classified in the _____ order. Mammalian
9. Schaffhausen first found Neanderthal fossils in _____ in the Neander Valley. 1856
10. African apes are sometimes placed together in the family _____. Hominidae
11. Homininae members are called _____. Hominines
12. _____ seems to be one of the earliest of the major hominine characteristics to have evolved. Bipedalism
13. _____ is usually seen as the defining trait of the sub family. Bipedalism
14. _____ is the sophisticated human behavior. Brain body and size
15. Most modern human beings have a brain case volume of between _____ and _____ cm³. 1200, 1500
16. The bones suggest that females may have been _____ to _____ m in height and about _____ to _____ kg in weight. 0.9, 1.2, 27, 32
17. The bones suggest the males may have been some what more that _____ m tall, weighing about _____ kg. 1.5, 68
18. Australopithecus is about _____ years ago. 3.2 million
19. Homo sapiens sapiens is about _____ years ago to present. 100,000
20. Homo erectus is about _____ years ago. 750,000
21. The human skull has changed dramatically during the past _____ years. 3 million
22. _____ also theorize that the brain evolved a high capacity for learning and reasoning. Anthropologists
23. The evolutionary transition to homo probably occurred between about _____ and _____ years ago. 2.3 million, 2.7 million
24. A number of fossil skulls and jaws has been placed in the category Homo habilis meaning _____. Handy man
25. The earliest of stone tools come from the sites in Africa dated to about _____ years ago. 2.5 million
26. Fossil evidence of a large brained, small toothed form, from northern Kenya and _____ to _____ years ago. 1.8 million, 1.9 million
27. _____ is the first part of the time span of homo erectus was apparently limited to eastern African. Workman
28. Homo erectus is also known as _____. Homo ergaster
29. _____ is the dispute over the Neanderthals also involves the question of the evolutionary origin of the modern population. Races
30. Plant and animals began to be domesticated about _____ year ago. 12,000

LECTURE 5

1. The planet earth has often been converted to a vast _____ Spaceship
2. Name 5 things living things aboard the space craft do. Produce oxygen, cleans the air, adjust it gases, transfer energy, recycle waste product with great efficiency.
3. _____ deals with the non-living environment as well as the living. Science of ecology
4. _____ is also called the greatest system of life. Ecosphere
5. The earth is composed of the _____, _____, _____, _____ Biosphere, Hydrosphere, Atmosphere, Lithosphere.
6. _____ wraps the core of the earth Soil mantle
7. _____ is the mixture of gases surrounding the earth Atmosphere
8. _____ has a gravitational field strong enough to prevent the gas from escaping. Atmosphere
9. The principal constituent of the atmosphere of the earth are _____ (____%) and _____ (____%) Nitrogen 78%, Oxygen 21%
10. The water vapor content of the air varies considerably, depending on the _____ and _____. Temperature, Relative humidity
11. With 100% relative humidity, the water vapor of the air varies from _____ to _____. 190 PPM, 42,000 PPM
12. The strong possibility exist that increase in atm CO₂, mainly is a result of fossil fuel combination over the past century may affect the earth's climate through the process known as _____. Greenhouse effect
13. _____ is the lowest layer of the earth atmosphere. Troposphere
14. _____ is the layer in which most clouds occur Troposphere
15. _____ is above the troposphere Stratosphere
16. _____ is characterized by a marked decrease in temperature as the altitude increases. Mesosphere
17. _____ is the region beyond the ionosphere. Exosphere
18. _____ is the outer limit of atmosphere Exosphere
19. The density of dry air at sea level is about _____ the density of water 1/800
20. Pressure is measured by a _____ Barometer
21. Normal atmospheric pressure at sea level is _____ 760 mmHg
22. Troposphere and most of the stratosphere can be explored by means of _____ Sounding balloons
23. Methane level have risen 11% since _____ 1978
24. The temperature as a rule of troposphere decreases upward as the rate of _____ per _____ 5°C, 1000m
25. The study of air samples shows that at least _____ km above the sea level. 88
26. The troposphere extends to about _____ in tropical region 16km
27. _____ mostly occur in clouds. Troposphere.
28. The layer from 50 to 80 km is called _____ Mesosphere
29. The ozone became the subject of concern in the early _____ 1970s
30. CO₂ increased to _____ for the past 100 years 350ppm

LECTURE 6

1. _____ is the capacity of a physical system to perform work Energy
2. _____ possesses energy as a result of its motion or its position in relation to forces acting on it. Matter
3. _____ possesses energy in relation to its wave length and frequency Electromagnetic radiation
4. Energy associated with motion is known as Kinetic energy
5. Energy related to position is called Potential energy
6. List 5 types of energy. Chemical energy, Electrical energy, Solar energy, Mechanical energy, Nuclear energy
7. In the process of transformation, either _____ or _____ energy may be gained or lost Kinetic, Potential
8. A weight suspended from a chord has a potential energy due to its _____ Position
9. A piece of magnesium as its potential energy stored in _____ form Chemical
10. If a gun is fired, the chemical potential energy of the gun powder is transformed into _____ of a moving projectile. Kinetic energy
11. The kinetic energy of a moving rotor of a dynamo is changed into electrical energy by _____ Electromagnetic induction
12. The conservation of energy constitutes one of the basic principles of _____ Classical mechanics
13. In modern physics the two concepts, the conservation of energy and mass are _____ Unified
14. What are the 2 fundamental forces in nature Gravitational and Nuclear
15. _____ is a source of solar energy. Nuclear fusion
16. Renewable resources have cycling time less than _____ years while non-renewable resources are greater than _____ year. 100, a million
17. The depletable resources are _____ Fossil fuels
18. List 5 examples of renewable resources. Hydro energy, Solar energy, Wind, Biomass, biogas
19. The renewable solar energy is subdivided into _____ and _____ types. Direct, Indirect
20. The commercial source of energy is _____ Natural

21. Give 3 examples of commercial and non-commercial source of energy.
Non-commercial: Firewood, Animal dung, Biomass
Commercial: Coal, Oil & Nuclear source
22. Give 3 examples of conventional and non-conventional source of energy.
Non-conventional: Solar, Wind, Biogas.
Conventional: Coal, Oil, Hydro
23. Give 3 examples of primary and secondary source of energy.
Electrical, Mechanical.
Primary: Coal, Fire wood. Secondary:
24. ____ is the energy present in the natural flow of wind, water and sunlight in the environment. Renewable energy
25. The total solar resources are over ____ times. The world's current annual primary energy use. 10,000
26. All forms of energy are ____ by appropriate possesses.
27. ____ is a natural resource which will replenish to replace the portion depleted by usage and consumption. Renewable energy
28. Renewable energy is often referred to as ____ Clean energy
29. Solar or ____ cells are made from silicon or other materials that transform sunlight directly into electricity. Photovoltaic (PV)
30. Floating solar farms or ____ can be an effective use of waste water facilities and bodies of water that aren't ecological sensitive. Photovoltaic
31. According to the ____, more energy from the sun falls on the sun falls on the earth in one hour than is used by everyone in the world in one year. Renewable Energy Laboratory
32. ____ is organic material that comes from plant and animals and includes crops, waste wood and trees. Biomass
33. ____ energy uses the heat trapped in the earth's core which is created by the slow decay of radio active particles in rocks at the center of the planet. Geothermal energy
34. ____ is important as it has the potential to provide a ready supply of power without using natural resources. Renewable energy
35. Name the top 10 countries that has the highest amount of renewable energy with their percentage. Germany (12.74%), UK (11.95%), Sweden (10.96%), Spain (10.17%), Italy (8.8%), Brazil (7.35%), Japan (5.3%), Turkey (5.25), Australia (4.75), USA (4.32%)
36. Name the 4 major types of nonrenewable resources. Oil, Natural gas, Coal and Nuclear energy
37. ____ are formed within the earth from dead plants and animal over a million years. Fossil fuels
38. ____ is the hydrocarbon containing material formed underground from the remains of dead plants and animals that human extract and burn to release energy for use.

**IT IS NOT OVER UNTIL IT IS
OVER**

DREX KEEPS IT